```python
# ============================================================
# NAIVE BAYES CLASSIFICATION
# Confusion Matrix and Accuracy
# Google Colab Ready
# ============================================================

# Import required libraries
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report


# ------------------------------------------------------------
# 1. Load Dataset
# ------------------------------------------------------------

iris = load_iris()

X = iris.data
y = iris.target

print("Dataset Shape:", X.shape)
print("Classes:", iris.target_names)


# ------------------------------------------------------------
# 2. Split Dataset into Training and Testing Data
# ------------------------------------------------------------

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.30,
    random_state=42,
    stratify=y
)

print("\nTraining samples:", len(X_train))
print("Testing samples :", len(X_test))


# ------------------------------------------------------------
# 3. Create Naive Bayes Classifier
# ------------------------------------------------------------

model = GaussianNB()


# ------------------------------------------------------------
# 4. Train the Model
# ------------------------------------------------------------

model.fit(X_train, y_train)


# ------------------------------------------------------------
# 5. Make Predictions
# ------------------------------------------------------------

y_pred = model.predict(X_test)


# ------------------------------------------------------------
# 6. Calculate Accuracy
# ------------------------------------------------------------

accuracy = accuracy_score(y_test, y_pred)

print("\n===================================")
print("NAIVE BAYES RESULTS")
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```python
print("====================================")

print("\nAccuracy:", accuracy)
print("Accuracy (%):", accuracy * 100)


# ------------------------------------------------------------
# 7. Display Confusion Matrix
# ------------------------------------------------------------

cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix:")
print(cm)


# ------------------------------------------------------------
# 8. Plot Confusion Matrix
# ------------------------------------------------------------

plt.figure(figsize=(7, 5))

sns.heatmap(
    cm,
    annot=True,
    fmt="d",
    cmap="Blues",
    xticklabels=iris.target_names,
    yticklabels=iris.target_names
)

plt.title("Naive Bayes - Confusion Matrix")
plt.xlabel("Predicted Class")
plt.ylabel("Actual Class")
plt.show()


# ------------------------------------------------------------
# 9. Classification Report
# ------------------------------------------------------------

print("\nClassification Report:")
print(
    classification_report(
        y_test,
        y_pred,
        target_names=iris.target_names
    )
)
```

```
Dataset Shape: (150, 4)
Classes: ['setosa' 'versicolor' 'virginica']

Training samples: 105
Testing samples : 45

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NAIVE BAYES RESULTS
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Accuracy: 0.9111111111111111
Accuracy (%): 91.11111111111111

Confusion Matrix:
[[15  0  0]
 [ 0 14  1]
 [ 0  3 12]]
```

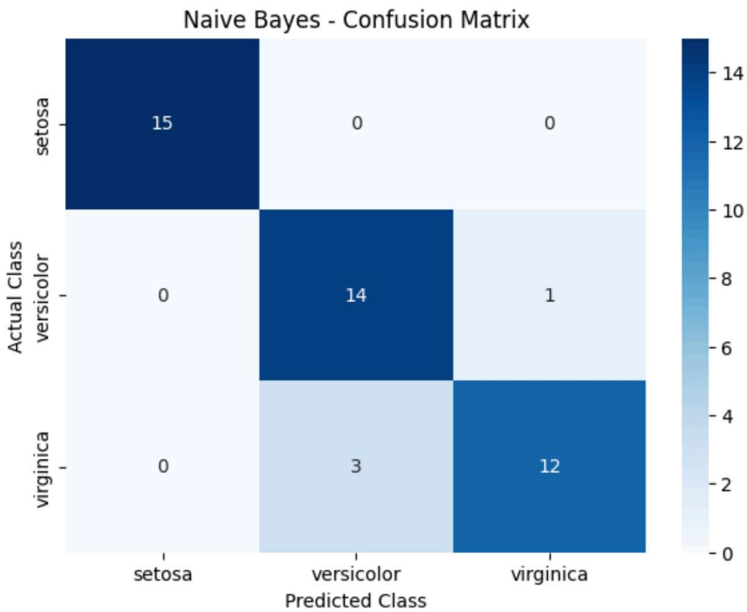


```
Classification Report:
              precision    recall  f1-score   support

      setosa       1.00      1.00      1.00        15
  versicolor       0.82      0.93      0.88        15
```